



CHAPTER 4

Post-Supernova Kinematics: The Natal Kick



And the memories were lost long ago

But at least you have beautiful ghosts.

— *Taylor Swift* —



THE TERMINUS OF A MASSIVE STAR’S life is a core-collapse supernova, when the hydrostatic equilibrium between the pressure of the star and its self-gravity is broken, and the stellar core collapses in on itself. Once the star reaches nuclear densities, neutron degeneracy is able to stabilise the remnant, causing a shockwave to propagate outwards from the core. The energy released is $\sim 10^{53}$ erg, of which $\sim 10^{51}$ erg is transferred to the star (Eldridge & Tout, 2018), resulting in the ejection of an amount of mass we denote M_{ej} and leaving a remnant of mass M_{rem} behind. If the ejection is asymmetric, the collapse endows the nascent remnant with a velocity v_k whose overall direction is antialigned to the more massive ejecta. This motion is termed a natal supernova kick.

Given an observation of the proper velocity of a compact remnant, it is in principle possible to trace its motion back and ascertain its kick velocity. However, this is not necessarily a straightforward endeavour. If the proper velocity is high and the remnant old, it has travelled quite a distance from its explosion site to be located at the point the observation has been taken. In addition, the star has potentially had its proper velocity modified by gravitational attraction from either other runaway stars or its host galaxy’s potential. One would require the exact location in 3D space as well as the 3D velocity and remnant age in order to ascertain the motion, but from there the inverse problem becomes straightforward.

4.1 | Overview

As a consequence of conservation of momentum, it can be shown that if the ejecta is assumed to eject along a preferred axis, the velocity of the kick can be written as

$$v_k = \alpha \left(\frac{M_{\text{ej}}}{M_{\text{rem}}} \right) + \beta \left(\frac{M_{\text{NS}}}{M_{\text{rem}}} \right), \quad (4.1)$$

for free parameters α and β (both measured in km s^{-1}). Here, M_{NS} is the mass of a NS; if the kick is applied to a NS, then $M_{\text{NS}} = M_{\text{rem}}$ and the coefficient reduces to unity. If the kick is applied to a BH, then we can take $M_{\text{NS}} = 1.4 M_{\odot}$, which serves to convert the distribution from one in velocity-space to one in momentum-space; see Bray & Eldridge (2016, 2018); Richards et al. (2023) for further detail.

This parameterization introduces, effectively, four free parameters in $\alpha, \beta, \theta, \phi$ (the latter two being azimuthal and equatorial angles of the ejecta mass). We assume that the preferential axis is sampled uniformly across the 4π steradians of the star, and that our only two free parameters are α and β . It is pertinent to note two things due to this assumption: firstly, we are not assuming that the ejection occurs along one axis and that the kick is aligned along a different, random, axis. Rather, we are assuming that the asymmetric ejection occurs along a randomly sampled axis and that a positive-velocity kick is antialigned to that. The second note, explored further in Section 4.1.2, is that there is considerable uncertainty around the nature of the kick and the precise mechanism that causes it to eject along a given axis. This is the reason we sample the kick direction uniformly, though we will simulate sufficiently many systems (on the order of $\mathcal{O}(10^{12})$) that this will be marginalised over.

The existence of two free parameters, α and β , mean that a choice on the values must be made. The

“traditional” way to choose these parameters, as outlined in [Bray & Eldridge \(2016, 2018\)](#), is to refine them based on isolated pulsar velocities. This methodology, however, has significant drawbacks. Notably, the reliance on one observation can engender biases in the result at the expense of no longer predicting accurate results for other observables. For instance, in the limit of ultra-stripped supernovae (USSNe) where the ejecta-remnant mass ratio becomes small, the fiducial kick in [Bray & Eldridge \(2018\)](#) would produce kick velocities that are strongly negative (indicating a strong kick aligned with the direction of motion of the ejecta), contrasted with the weak kicks we expect from these supernovae (e.g., [Willcox et al., 2021](#)). It becomes evident that a new series of empirical tests are needed, whereupon more constraints than simply pulsar velocities are tested. In this Chapter, we outline four such tests that form the basis of the work of [Richards et al. \(2023\)](#). These tests – using observed LVK Collaboration merger rates, period-eccentricity distributions of Binary Neutron Stars (BNSs), isolated pulsar velocities, and production of USSNe – allow us to narrow the parameter space significantly, with the promise of future refinement with more observations (such as through LVK Collaboration O4).

4.1.1 | Parameter space

We perform our computation on three populations, termed FINEGRID, WIDEGRID, and HOBBS, whose properties are outlined in Table 4.1. These populations simulate 1,000 kicks per supernova in each stellar evolution model. The aim of the three populations is twofold. Firstly, by generating FINEGRID and WIDEGRID, we are able to look both at fine detail in the region at which the most likely value is expected to lie and also the large-scale behaviour over wider parameter ranges.

Population Name	α_{low} [km s ⁻¹]	α_{high} [km s ⁻¹]	β_{low} [km s ⁻¹]	β_{high} [km s ⁻¹]	Step Size [km s ⁻¹]
FINEGRID	0	400	-200	200	10
WIDEGRID	0	700	-350	350	50
HOBBS	-	-	-	-	-

Table 4.1: The populations under study. HOBBS is a fiducial kick prescription as given by [Hobbs et al. \(2005\)](#), which is a Maxwellian distribution with distribution parameter $\sigma = 265 \text{ km s}^{-1}$, based on an empirical fit to the velocities of 233 pulsars. All three prescriptions simulated 1,000 kicks for each system, with the resultant kick being the mean velocity sampled, and the direction of the kick being sampled uniformly across the 4π steradians of the orbit.

Secondly, by using [Hobbs et al. \(2005\)](#), we are able to create a population governed by a kick that has been well-tested in literature (see, for example, [Bouffanais et al., 2021](#); [Broekgaarden et al., 2021](#); [Fragione, 2021](#); [Igoshev et al., 2021](#); [Mandel & Broekgaarden, 2022](#)) which serves as a fiducial comparison. From a Bayesian perspective, we are essentially choosing the HOBBS distribution as our prior for the express purpose of a Bayesian test in Section 4.2.2.

The choice of extremal values for α and β are somewhat arbitrary. Figure 4.2 shows a wider parameter space with $\alpha \in [0, 1000]$, $\beta \in [-500, 500]$. Regions in the upper right (bottom left) corner of the plot have quite high (low) mean kick velocities. Although both grids encroach on these regions somewhat, there is

a tradeoff between exploring the grid space fully and using computational resources efficiently. Our mean kick velocities should not be too extreme; the mean for a [Hobbs et al. \(2005\)](#) kick is $\sigma = 265 \text{ km s}^{-1}$, for a [Verbunt et al. \(2017\)](#) is $\sigma_1 = 75 \text{ km s}^{-1}$ and $\sigma_2 = 316 \text{ km s}^{-1}$, and for the observed [Willcox et al. \(2021\)](#) kick $\sigma_1 \approx 30 \text{ km s}^{-1}$, $\sigma_2 \approx 230 \text{ km s}^{-1}$. These¹ are shown on the grid as black lines.

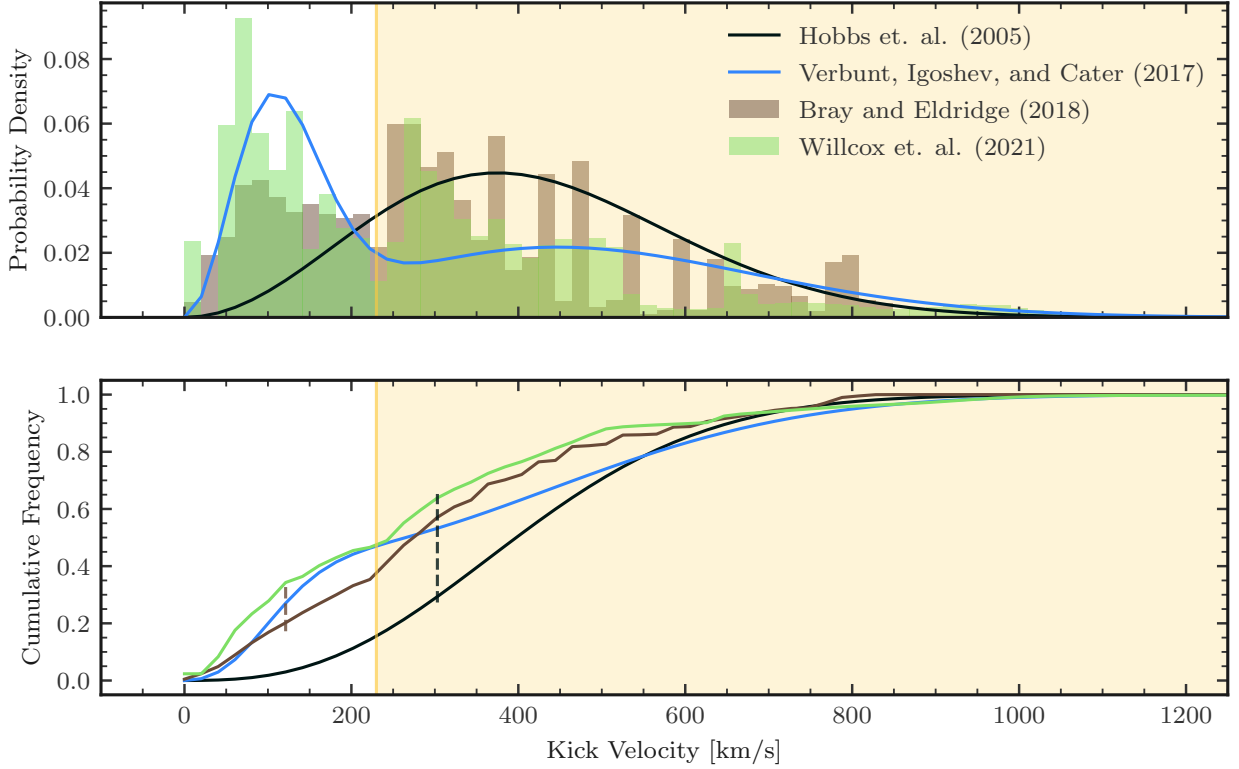


Figure 4.1: The Probability Density Function (PDF) and Cumulative Distribution Function (CDF) of the distributions using kicks from [Hobbs et al. \(2005\)](#); [Verbunt et al. \(2017\)](#); [Bray & Eldridge \(2018\)](#); [Willcox et al. \(2021\)](#). [Hobbs et al. \(2005\)](#) has a closed-form expression, whereas the others are computed empirically, hence the Bray and Willcox CDFs are less smooth than that of Hobbs. The two dashed lines indicate where and how the Kolmogorov-Smirnov (KS) statistic is measured for the associated distribution. The yellow region indicates the region for which $v_{\text{kick}} \geq v_{\text{TGRV}}$, the transverse Galactic rotation velocity of the Sun, $\sim 230 \text{ km s}^{-1}$.

Whilst the exact parameters can be shifted somewhat and maintain the statistical integrity given these constraints, there are a few additional heuristic constraints we place on the regions to select. First, we wish to capture low-velocity USSN kicks (as will be discussed in Section 4.2.4), necessitating a region near $\alpha = \beta = 0$ (where the kick would be identically zero for all ejecta-remnant mass ratios). Secondly, the fiducial work of [Bray & Eldridge \(2018\)](#) found $\alpha = 100, \beta = -170$. Although a goal was to (ideally) remove $\beta < 0$ as a possible value (for USSN again), to impose that as a prior when the fiducial value sat within that range would have gone contrary to typical Bayesian practices. Thirdly, we expected but could not guarantee that the magnitude of each parameter would shrink with further tests, given the large uncertainty bands that [Bray & Eldridge \(2018\)](#) found (hence broadening our search area beyond $|\alpha| = 100, |\beta| = 170$). Finally, we elected to increase the grid search region by 75 per cent for WIDEGRID as this would allow us to just reach $\bar{v}_k = 700 \text{ km s}^{-1}$, which is where the CDF of the [Hobbs et al. \(2005\)](#); [Verbunt et al. \(2017\)](#); [Willcox et al. \(2021\)](#) distributions

¹Due to the grid size and the random sampling, a true contour is difficult to observe due to sampling noise. The contours shown are thus linear fits to the actual contours.

all approximately begin to level out (signifying few velocities above those), as shown in Figure 4.1.

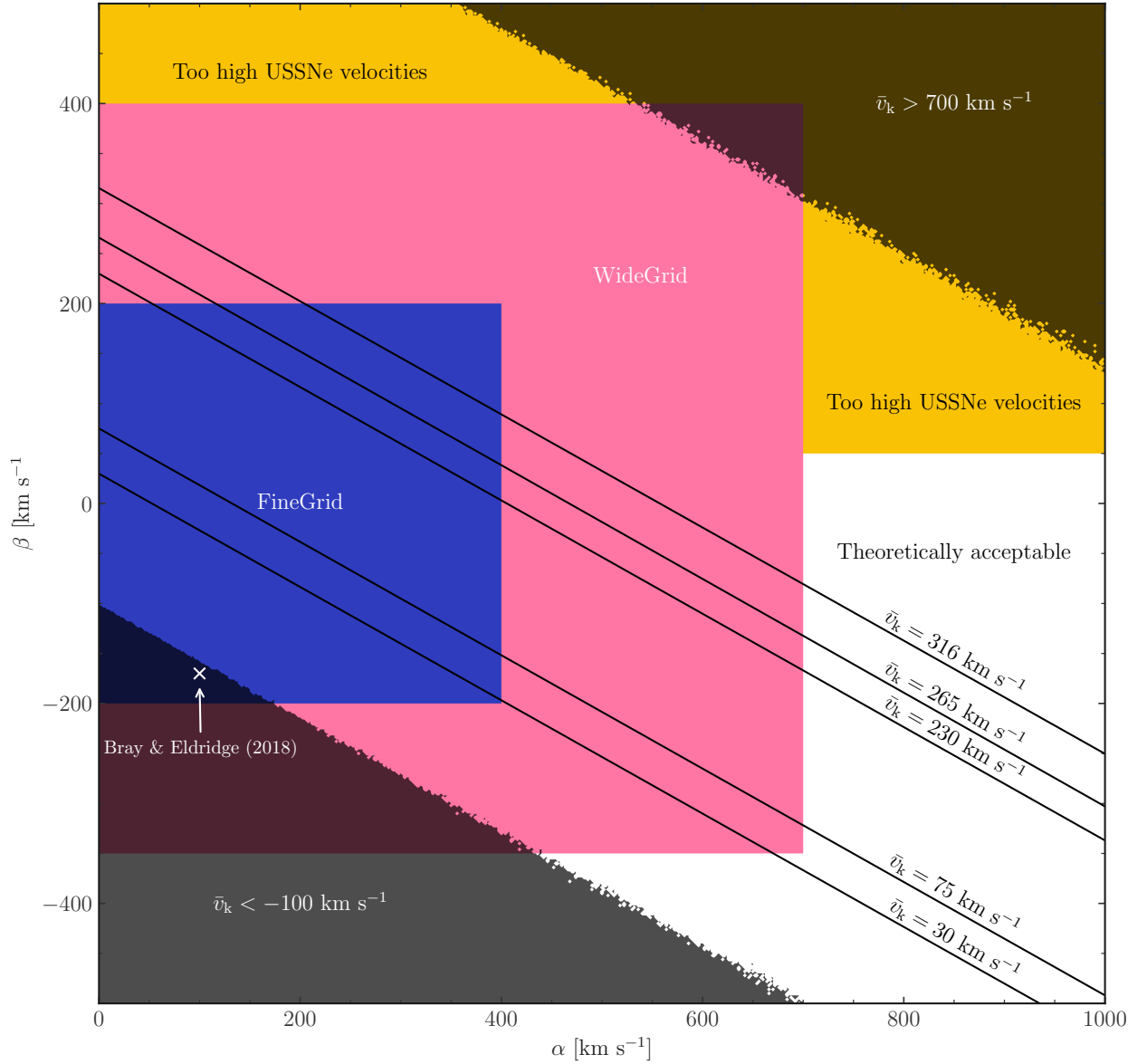


Figure 4.2: A schematic of how FINEGRID and WIDEGRID are related. The mean kick velocity v_k was computed by averaging the kick velocities of 1,000 simulated kicks for each combination of α and β , with $m_{\text{rem}} \in [0.1, 2.5]$ and $m_{\text{ej}} \in [0.1, m_{\text{rem}}]$ sampled uniformly. The white cross represents the point $\alpha = 100, \beta = -170$, which are the fiducial kick parameters from [Bray & Eldridge \(2018\)](#).

4.1.2 | Natal Kick Prescriptions

Cats can disappear.

— FOURTH OF THE SIX IMPOSSIBLE THINGS

There are conflicting accounts as to the cause of the kick (see, for example, [Fryer et al., 1998](#); [Arzoumanian et al., 2002](#); [Bombaci & Popov, 2004](#); [Hobbs et al., 2005](#); [Bray & Eldridge, 2016](#); [Verbunt et al., 2017](#)). There are conflicting accounts as to the form and distribution of the kick too, as discussed in, for example, [Blaauw \(1961\)](#); [Gunn & Ostriker \(1970\)](#); [de Loore et al. \(1975\)](#); [Sutantyo \(1978\)](#); [Hobbs et al. \(2005\)](#); [Baker et al.](#)

(2008); Wongwathanarat et al. (2013); Bray & Eldridge (2018); Giacobbo & Mapelli (2020); Mandel & Müller (2020).

We stress that we are not attempting to prescribe *which* prescription is somehow “best”, but seek an easy-to-configure kick which allows us to configure kicks without changing any of the physics of how our population was formed. This can be done by modifying the kick of Bray & Eldridge (2016), which takes two free parameters which can be changed without imposing constraints on physics. This prescription is hereinafter referred to as the “Bray kick”. Similarly, we require for the purposes of computing log-Likelihoods in Section 4.2.2 a comparison kick, for which we select the kick of Hobbs et al. (2005), a well-known empirically derived kick. This prescription is hereinafter referred to as the “Hobbs kick”.

The Bray kick velocity is prescribed by Eqn(4.1). When applied to a NS, M_{NS} is by definition the remnant mass, so the kick velocity becomes

$$v_k = \alpha \left(\frac{M_{\text{ej}}}{M_{\text{rem}}} \right) + \beta. \quad (4.2)$$

On the other hand, the Hobbs kick velocity is given by the PDF

$$P(v_k) dv_k = \sqrt{\frac{2}{\pi}} \frac{v_k^2}{\sigma^3} \exp \left(-\frac{v_k^2}{2\sigma^2} \right) dv_k, \quad (4.3)$$

where $\sigma = 265 \text{ km s}^{-1}$ is the distribution parameter found by fit in Hobbs et al. (2005) to the velocities of 233 pulsars.

4.1.3 | The TAKAHE orbital evolution code

As part of this work, we have developed the TAKAHE² compact object binary orbit evolution code, which solves the coupled equations of motion Eqn(2.25) and Eqn(2.26). Originally, to facilitate the equations being solved rapidly, we wrote the code as a PYTHON wrapper to a JULIA (Bezanson et al., 2017) integrator. However, the PYJULIA library (Arakaki et al., 2023) proved to be fragile³ and is not in fact supported officially on recent Python versions. To facilitate the potential interoperability between KAITIAKI and TAKAHE (and to remove a prolonged loading time when the JULIA runtime was loaded, given that PYJULIA precompiles our modules), we elected to rewrite the core integrator into PYTHON and to optimise it using traditional PYTHON methods instead.

There are two relevant functions in TAKAHE. The remaining functions are utility functions that allow us to (for instance) convert between lookback time / redshift, and period / separation. These functions are `integrate_eoms()` (Code Sample F.1), which solves the coupled differential equations for $\frac{da}{dt}$ and $\frac{de}{dt}$ (Eqn(2.25) and Eqn(2.26)), and `integrate_timescale()` (Code Sample F.2), which solves Eqn(2.29) for the coalescence time of the binary. With sufficient performance optimisations,⁴ `integrate_timescale()` is able

²The Takahē is an endangered New Zealand bird, which is viewed as taonga to the Ngāi Tahu iwi. There are approximately 500 left, all in conservation areas in the Te Waipounamu / the South Island.

³It requires custom installation procedures (so cannot be included as a dependency in e.g. a PYTHON package), would hang with no discernible output or ability to ascertain why, and could not handle KeyboardInterrupt exceptions in PYTHON.

⁴Primarily, just-in-time compilation with the NUMBA (Lam et al., 2015) library.

to achieve speeds of approximately 18,000 numerical integrations per second on an office PC⁵ on a sample of Neutron Star - Black Hole Binary (NS-BH) binaries in the LISA dataset (Tang, priv. comm.).

Algorithmically, both codes employ a simple Euler integration routine (where $f_{n+1} = f_n + \frac{df_n}{dt} dt$) and the choices work as follows:

1. When `integrate_eoms()` takes a step in time, it evaluates da and de . It then computes the relative changes in both e and a that arise by taking this timestep (through $\delta f = \left| 0.01 \frac{f}{f'} \right|$), and takes a timestep corresponding to the smallest change between a and e , or half of the width of the smallest BPASS time bin.
2. `integrate_timescale()` breaks up Eqn(2.29) into four cases dependent on the initial eccentricity. Let τ be the coalescence time of a circular binary ($\tau = \frac{a_0^4}{4\beta}$), then the cases are:

$$t_c = \begin{cases} \tau & e_0 = 0 \\ \frac{c_0^4}{4\beta} e_0^{\frac{48}{19}} & e_0 < 0.01 \\ \frac{12}{19} \frac{c_0^4}{\beta} \int_0^{e_0} \frac{e^{\frac{22}{19}} \left(1 + \frac{121}{304} e^2\right)^{\frac{1181}{2299}}}{(1-e^2)^{\frac{3}{2}}} de & 0.01 \leq e_0 \leq 0.99 \\ \tau^{\frac{768}{425}} (1-e_0^2)^{\frac{7}{2}} & 0.99 < e_0 \end{cases}, \quad (4.4)$$

where all symbols have the same meaning as in Section 2.5. The integral in the intermediate case is evaluated numerically, taking a fixed timestep of $de = \frac{e_0}{1000}$ such that the discretised sum becomes nothing more than

$$t_c (0.01 \leq e_0 \leq 0.99) = \frac{e_0}{1000} \frac{12}{19} \frac{c_0^4}{\beta} \sum_{e=0}^{e_0} \frac{e^{\frac{22}{19}} \left(1 + \frac{121}{304} e^2\right)^{\frac{1181}{2299}}}{(1-e^2)^{\frac{3}{2}}}, \quad (4.5)$$

and dependence of the coalescence time t_c on the initial separation a_0 is handled through the c_0 term. Since the step size is an arbitrary choice, a tradeoff can be made between accuracy and runtime. Figure 4.3 shows the effect of changing the step size ($\frac{e_0}{1000}$ in Eqn(4.5)) on convergence to a final result, and Figure 4.4 shows the runtime as a function of step size on an office computer.

4.2 | Observational Constraints

In this section, we outline the four tests we use to refine the acceptable parameters of the Bray kick.

4.2.1 | Merger Rates of BNS systems

The first test is to match the observed merger rates of BNS systems. To do this, we use the LVK Collaboration catalogue of results from Operating Run 3 (O3). The simplest test is to match the merger rate today, which we denote as $\mathcal{R}_0$, to that observed by LVK Collaboration (LIGO-Virgo, 2021; LIGO-Virgo-KAGRA,

⁵A Dell OptiPlex 7480 AIO with a 12-core 3.1 GHz Intel Core i5-10500 CPU, running no other programs except for during the final few iterations.

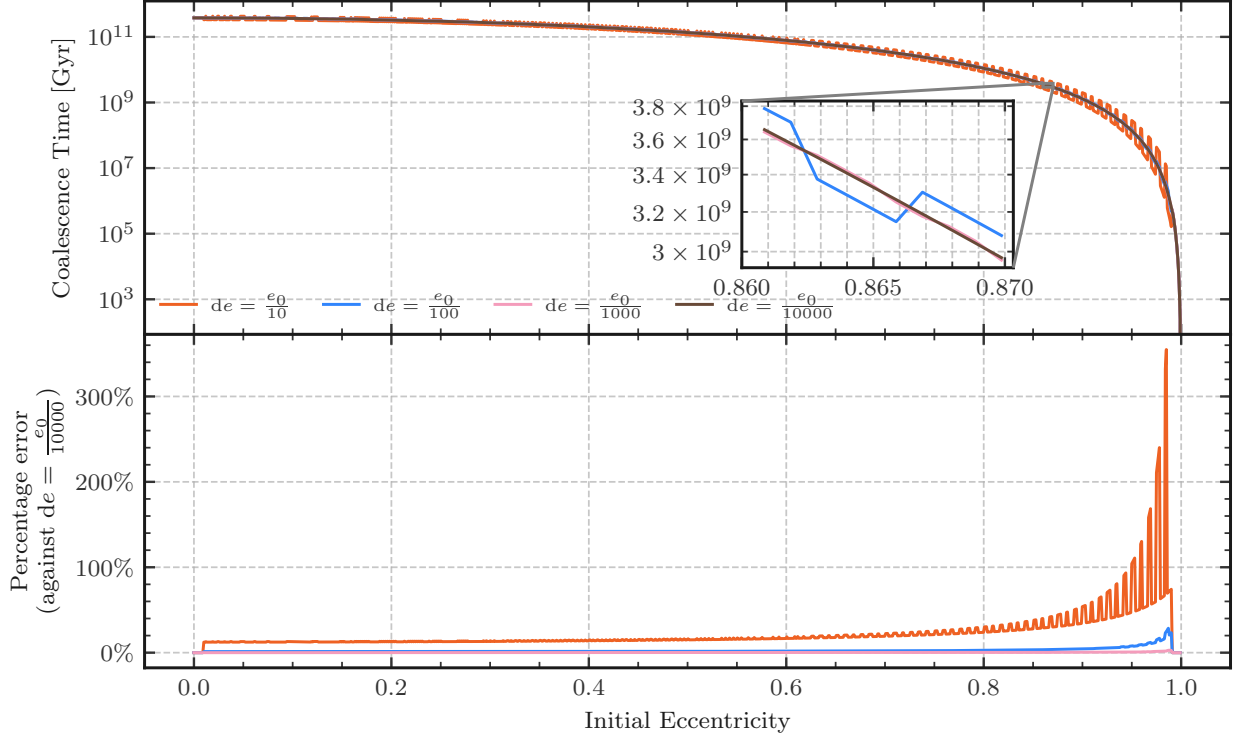


Figure 4.3: The effect of varying de in Eqn(4.5) on the coalescence time. In the inset plot, the $de = \frac{e_0}{10}$ line has been removed for clarity. This binary was assumed to have $m_1 = m_2 = 1.44 M_\odot$ and $a_0 = 1970.5 R_\odot$ (the mean separation of a sample of binaries with $\alpha = 100 \text{ km s}^{-1}$, $\beta = -170 \text{ km s}^{-1}$).

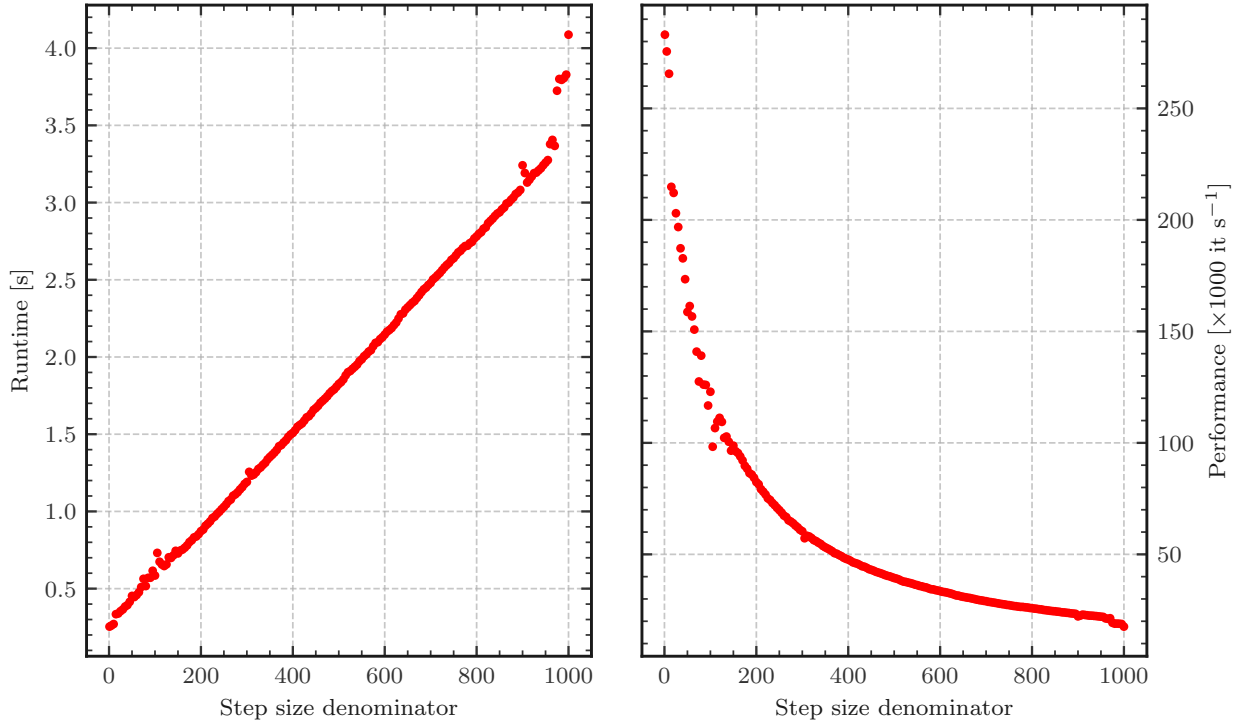


Figure 4.4: The effect of varying de in Eqn(4.5) on the runtime, which is demonstrably $\mathcal{O}(n)$, and the performance of the algorithm. The input data to this algorithm were the LISA binary sample from Tang (priv. comm.). The performance is shown in units of thousands of iterations per second.

2021). We display this in Figure 4.5 alongside the previously reported fiducial values of $\alpha = 100 \text{ km s}^{-1}$ and $\beta = -170 \text{ km s}^{-1}$, the theoretical merger rate calculation performed by Eldridge et al. (2019), and the observed merger rates from LIGO-Virgo (2021) and LIGO-Virgo-KAGRA (2021). We term the contour defined by LIGO-Virgo (2021) (in black) the “LIGO contour”.

Selecting the value to calibrate to is not a straightforward task. Although the LIGO-Virgo (2021) contour is superseded by the LIGO-Virgo-KAGRA (2021) contour, the separation into different inference models means that the merger rates for BNS systems have intractably large error bars covering the entire plane; for instance, their MULTISOURCE (MS) model alone spans $2.11 \leq \log \mathcal{R}_0 \leq 3.23$ which covers FINEGRID entirely and almost entirely covers WIDEGRID. In addition to the lack of constraint, we would be imposing additional parameters into the model insofar as the selection of model – whether, for example, PDB was taken as truth over BGP, MS, or one of the other population models. Whilst an interesting and assuredly insightful endeavour, the aim of this work is instead to develop a simple test that can be used in future on extra populations such as these. A simple criterion which can be extended is valuable, and future work may extend these tests in multiple directions.

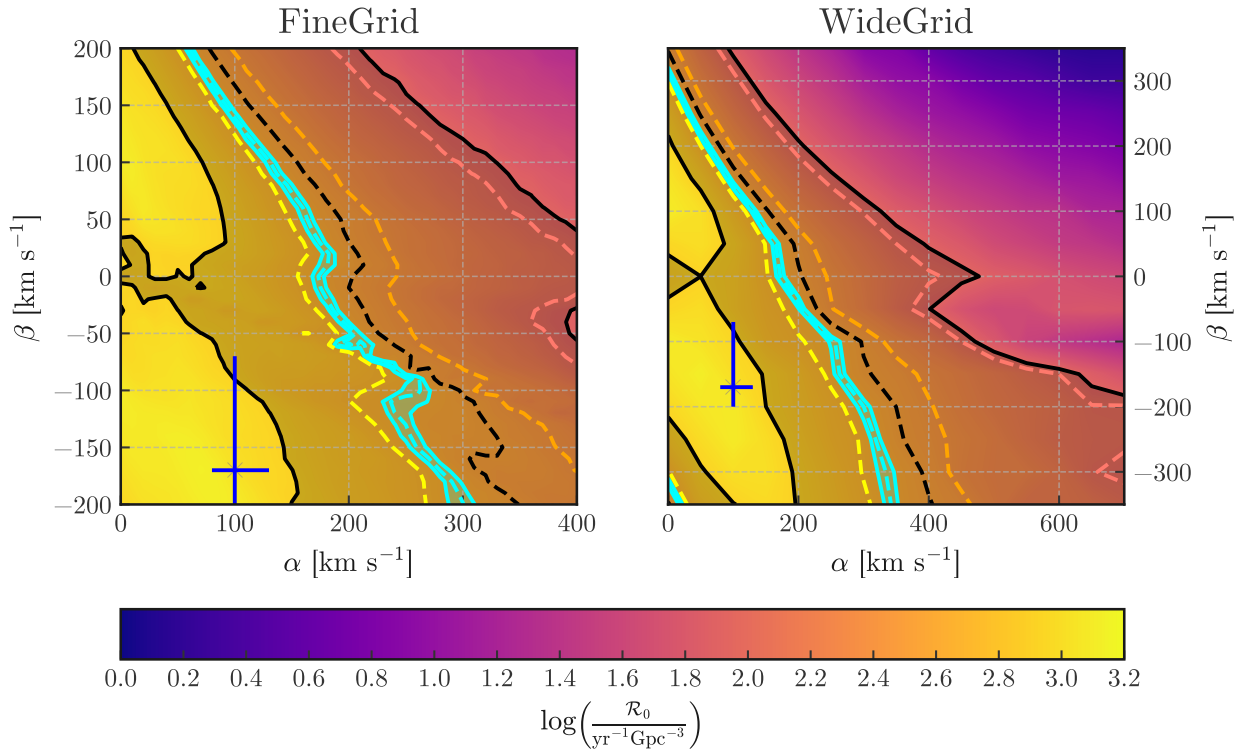


Figure 4.5: The merger rate as a function of kick parameters α and β . The cyan contour is the theoretical merger rate as reported by Eldridge et al. (2019). The black contour is the merger rate observed in LIGO-Virgo (2021). The remaining contours are the merger rates reported in LIGO-Virgo-KAGRA (2021) (pink: BGP, yellow: MS, orange: PDB [ind], see LIGO-Virgo-KAGRA (2021) for an explanation of each model). Shaded regions represent uncertainties, and uncertainties are not reported for the LIGO-Virgo-KAGRA (2021) results as they would cover the entire plane. The blue cross is the fiducial value of α and β from Bray & Eldridge (2018). This value lies off the previous work’s contours as the rejuvenation age calculation has been improved in TUI in preparation for the next major BPASS release.

4.2.2 | Period-Eccentricity Distributions

Since, in the absence of any mass transfer, the motion of two compact binaries is wholly dependent on their initial separation and orbital eccentricity, they trace out paths in a two-dimensional phase space. In general, since binaries circularise over time, and by the emission of gravitational waves their orbits shrink, a binary will start in the high-period, high-eccentricity region of period-eccentricity space, and evolve towards the low-period, low-eccentricity region. As a result, given a population of binary stars, we can design a density plot, where the density is effectively the likelihood of a binary selected at random having the parameters we sample.

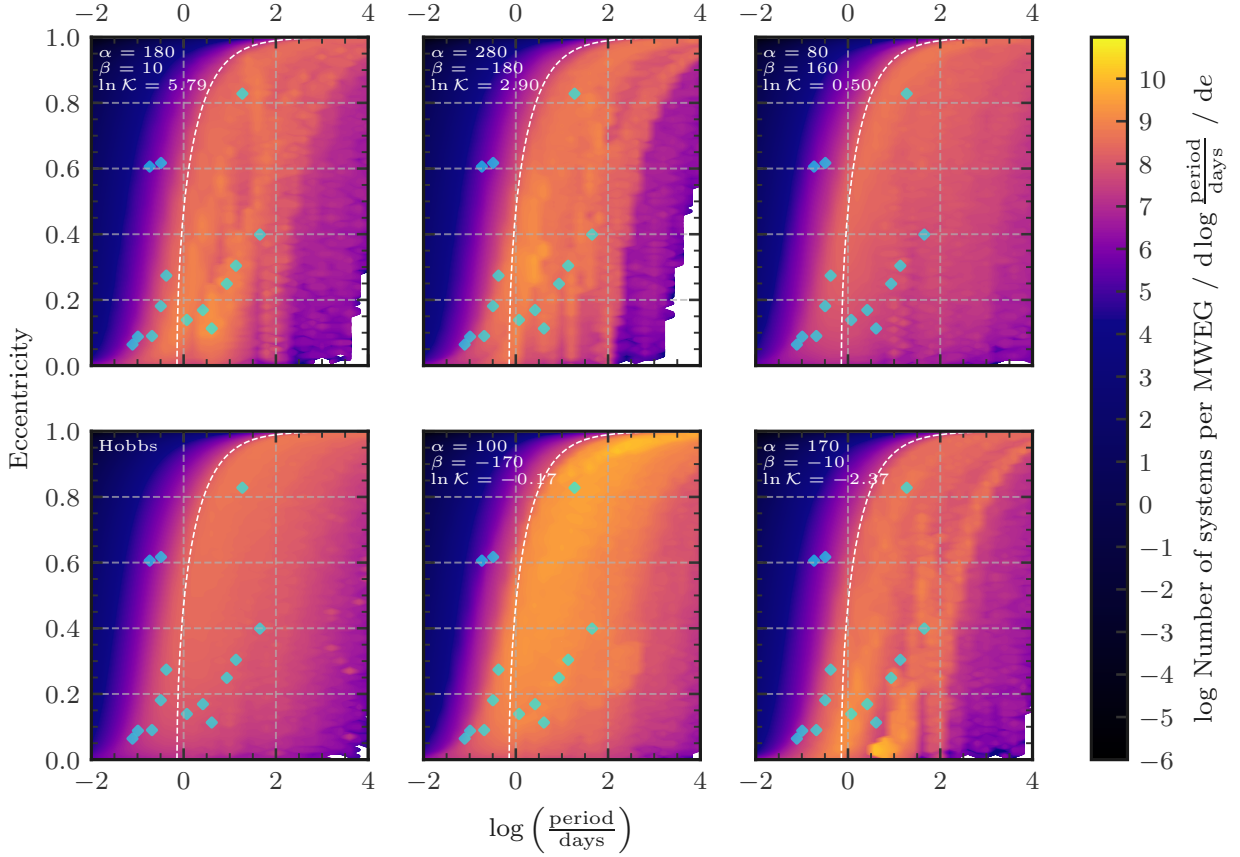


Figure 4.6: The period-eccentricity distributions for six selected [Bray & Eldridge \(2016\)](#) kicks. Shown are the kicks for $\alpha = 180 \text{ km s}^{-1}, \beta = 10 \text{ km s}^{-1}$ as the system with the highest $\ln \mathcal{K}$ across the LIGO contour, $\alpha = 170 \text{ km s}^{-1}, \beta = -10 \text{ km s}^{-1}$ as the lowest, and the balance to illustrate structural differences. White regions show regions where no systems evolve to. The white line represents a constant-time coalescence time of the Hubble time ($\sim 13.8 \text{ Gyr}$), computed using [Mandel \(2021\)](#). The cyan diamonds are the observed systems enumerated in Table 4.2. In white text are the values of (α, β) , and the $\ln \mathcal{K}$ values for each plot.

Computationally, for each binary in our sample, we trace its path through period-eccentricity space, and at every timestep we add $w \times dt$ to the density plot, where dt is the timestep, and w is the weight term output from the TUI code proportional to the IMF weight (where the constant of proportionality is related to the uniformly sampled kick direction). The result of this computation, for the parameter values that lie along the LIGO contour, are shown in Appendix C. We also show the same result in Figure 4.6. Given a catalogue of N (observed) binary stars, we can then compute the likelihood of the model as

Name	$\log(P/\text{days})$	e	Original Reference	Name	$\log(P/\text{days})$	e	Original Reference
J0453+1559	0.609808	0.113	Martinez et al. (2015)	J0737-3039	-0.991400	0.088	Kramer et al. (2006)
B1534+12	-0.375718	0.274	Fonseca et al. (2014)	J1756-2251	-0.494850	0.181	Faulkner et al. (2005)
B1913+16	-0.490797	0.617	Hulse & Taylor (1975)	J1913+1102	-0.686133	0.090	Lazarus et al. (2016)
J1757-1854	-0.735182	0.606	Cameron et al. (2018)	J1518+4904	0.936212	0.249	Janssen et al. (2008)
J1811-1736	1.273672	0.828	Corongiu et al. (2007)	J1829+2456	0.070407	0.139	Champion et al. (2004)
J1930-1852	1.653791	0.399	Swiggum et al. (2015)	J1753-2240	1.134751	0.304	Keith et al. (2009)
J1411+2551	0.417638	0.169	Martinez et al. (2017)	J1946+2052	-1.107905	0.064	Stovall et al. (2018)

Table 4.2: The systems we use in our study. These data were originally compiled in [Vigna-Gómez et al. \(2018\)](#).

$$\ln \mathcal{L} = \sum_{n=0}^N \ln p(\ln P_n, e_n), \quad (4.6)$$

where $p(\ln P_n, e_n)$ is the content of the density plot at the location of the n^{th} observed BNS system. Then, the Bayes factor (defined as the likelihood ratio) can be computed as

$$\mathcal{K} \equiv \frac{\mathcal{L}}{\mathcal{L}_{\text{Hobbs}}} \implies \ln \mathcal{K} = \ln \mathcal{L} - \ln \mathcal{L}_{\text{Hobbs}}, \quad (4.7)$$

where $\mathcal{L}_{\text{Hobbs}}$ is the log-Likelihood of the Hobbs kick. We use a catalogue of fourteen BNS systems originally compiled in [Vigna-Gómez et al. \(2018\)](#), shown in Table 4.2. From this catalogue, we sample the period-eccentricity distribution at that space and take that as the probability of observing that binary, thus allowing us to use the content of the period-eccentricity distribution as the $\ln p$ term in our summand of Eqn(4.6).

By doing this we effectively construct a Bayes-factor map, where $\ln \mathcal{K}$ can be shown evaluated at every combination of α and β . We term this value $\ln \mathcal{K}_{\alpha\beta}$. This is shown in Figure 4.7 for both of our considered grids. In general we see that approximately 60% of FINEGRID has a positive log-Bayes factor (indicating any level of support of the model compared to Hobbs), whereas approximately 40% of WIDEGRID has a positive log-Bayes factor. In a Bayesian analysis, the constraint that $\ln \mathcal{K} \geq 3.4$ indicates strong support for the model contrasted against the reference (i.e., [Hobbs et al., 2005](#)), which contains approximately 9.1% of FINEGRID and 4.4% of WIDEGRID.

4.2.3 | Single-Star Kick Velocities

The traditional methodology for refining a kick prescription is to consider the velocity distribution of isolated pulsars. Thus, given a suitable observational dataset, we would be able to compute the expected kick distribution and compare it, via a KS test, to the observed data set. The KS statistic between distributions $\mathcal{F}$ (the observed distribution) and $\mathcal{G}$ (the compared distribution) is given by

$$\text{KS} = \max |\text{cdf}(\mathcal{F}) - \text{cdf}(\mathcal{G})|, \quad (4.8)$$

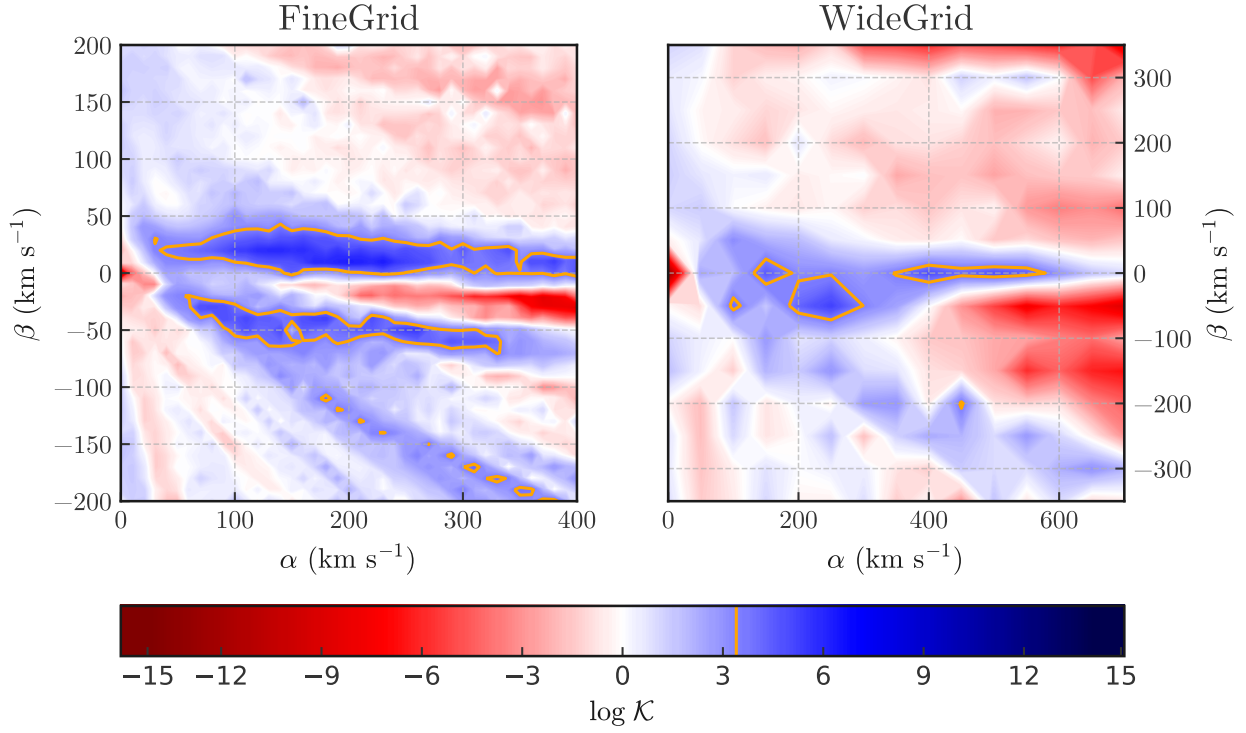


Figure 4.7: The $\ln \mathcal{K}$ values across $\alpha - \beta$ space. The orange line demarcates $\ln \mathcal{K} = 3.4$, which in a Bayesian test represents conclusive evidence in favour of the distribution compared to the [Hobbs et al. \(2005\)](#) kick.

where cdf means the CDF function, defined as the integral of the PDF. Given the CDF is bounded between 0 and 1 inclusive, the KS statistic is also bounded between this region. A KS statistic of 1 means that the compared distribution diverges perfectly from the observed distribution, whereas a KS statistic of 0 means it matches perfectly. This test of divergence was the refinement performed in [Bray & Eldridge \(2016, 2018\)](#), hence we select this (with an independently generated dataset to those works) to be consistent with prior work.

Pulsar velocity dataset

We use the same dataset of 81 pulsars from [Willcox et al. \(2021\)](#), and filter the dataset in the same way that they do. We remove millisecond pulsars, pulsars in globular clusters, and pulsars known to be in binaries, in order to homogenise the dataset. The PDF and CDF of [Willcox et al. \(2021\)](#), as well as those generated from [Hobbs et al. \(2005\)](#) and [Bray & Eldridge \(2018\)](#), is displayed in Figure 4.1.

This homogenised dataset gives the transverse velocities of each pulsar alongside the right ascension / declination, the proper motion in the right ascension / declination directions, and parallax. We assume that the transverse velocity is approximately the original kick velocity. As mentioned earlier in this chapter, old remnants with high velocities may have moved substantially – a 20 Myr remnant with a transverse velocity of 500 km s^{-1} will have moved $\sim 10 \text{ kpc}$, absent external forces ([Willcox et al., 2021](#)). However, this is likely only to be an issue for pulsars whose transverse velocity substantially exceeds the transverse Galactic rotational velocity of the Sun $v_{\text{TGRV}} \approx 230 \text{ km s}^{-1}$ ([Bhattacharjee et al., 2014](#)), shown in yellow on Figure 4.1. Although the dataset of [Willcox et al. \(2021\)](#) has pulsars with velocities higher than this value – peaking at $\sim 850 \text{ km s}^{-1}$

– this is not a grave concern for this work, for two reasons. Firstly, the kick velocities output from TUI are taken immediately after the supernova. Secondly, the analysis Willcox et al. (2021) performs is focused on low-velocity pulsars, and this test limits the kick velocity of an USSNe to 30 km s^{-1} , well below v_{TGRV} .

The Hobbs et al. (2005) kick has a closed-form expression for the PDF $P(v_k)$, given by Eqn(4.3). Consequently, it has a smooth curve in Figure 4.1, whereas the Bray & Eldridge (2018) and Willcox et al. (2021) PDFs are presented as histograms as they are computed.

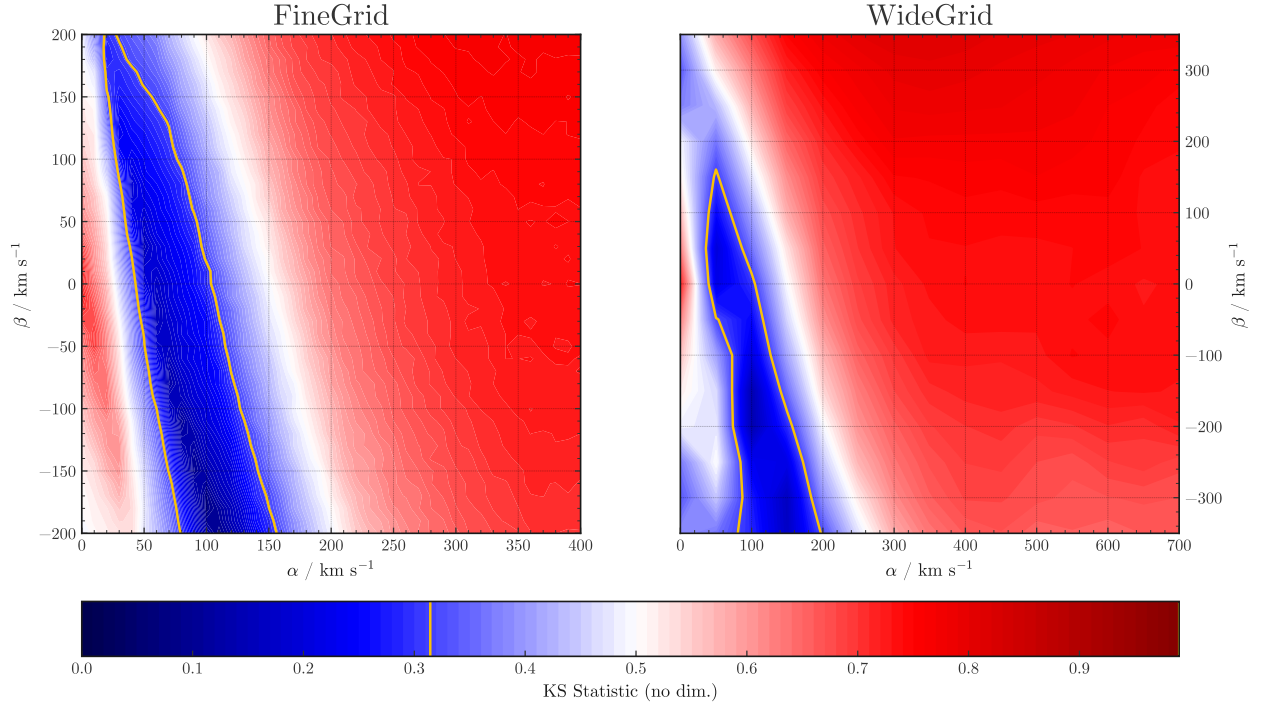


Figure 4.8: The KS statistic for FINEGRID and WIDEGRID. The orange line denotes the KS statistic for the Hobbs kick, which has a value of 0.34.

Comparison

Figure 4.8 shows the KS statistic as a function of α and β across our two parameter spaces. We also show the KS statistic of the Hobbs kick, which has a value of 0.34. We find in general a valley of models with good agreement to Willcox et al. (2021), and find that the global minimum KS statistic is at $\alpha = 100, \beta = -180$ – agreeing with the prior work of Bray & Eldridge (2018) (who find $\alpha = 100, \beta = -170$).

We also show in Figure 4.9 a selection of four PDFs, alongside their associated CDF and the CDFs for Willcox et al. (2021) and the Hobbs kick. We note that the numerical values shown in Figure 4.9 differs somewhat from those shown in Richards et al. (2023). This was due to a minor bug in the code incorrectly labelling the KS statistics for the figure. The actual result of the paper does not change with the correction of the bug, however the values reported in Figure 5 of Richards et al. (2023) do change.

The four values we present on the left side of Figure 4.9 correspond to $\alpha = 100 \text{ km s}^{-1}, \beta = -170 \text{ km s}^{-1}$ for the fiducial Bray kick, $\alpha = 180 \text{ km s}^{-1}, \beta = 10 \text{ km s}^{-1}$ to represent the system with the highest $\ln \mathcal{K}$ value from Section 4.2.2, $\alpha = 170 \text{ km s}^{-1}, \beta = -10 \text{ km s}^{-1}$ the lowest, and $\alpha = 40 \text{ km s}^{-1}, \beta = 0 \text{ km s}^{-1}$ the system with low α and zero β , corresponding to a kick that would *a priori* generate logical velocities in the USSNe

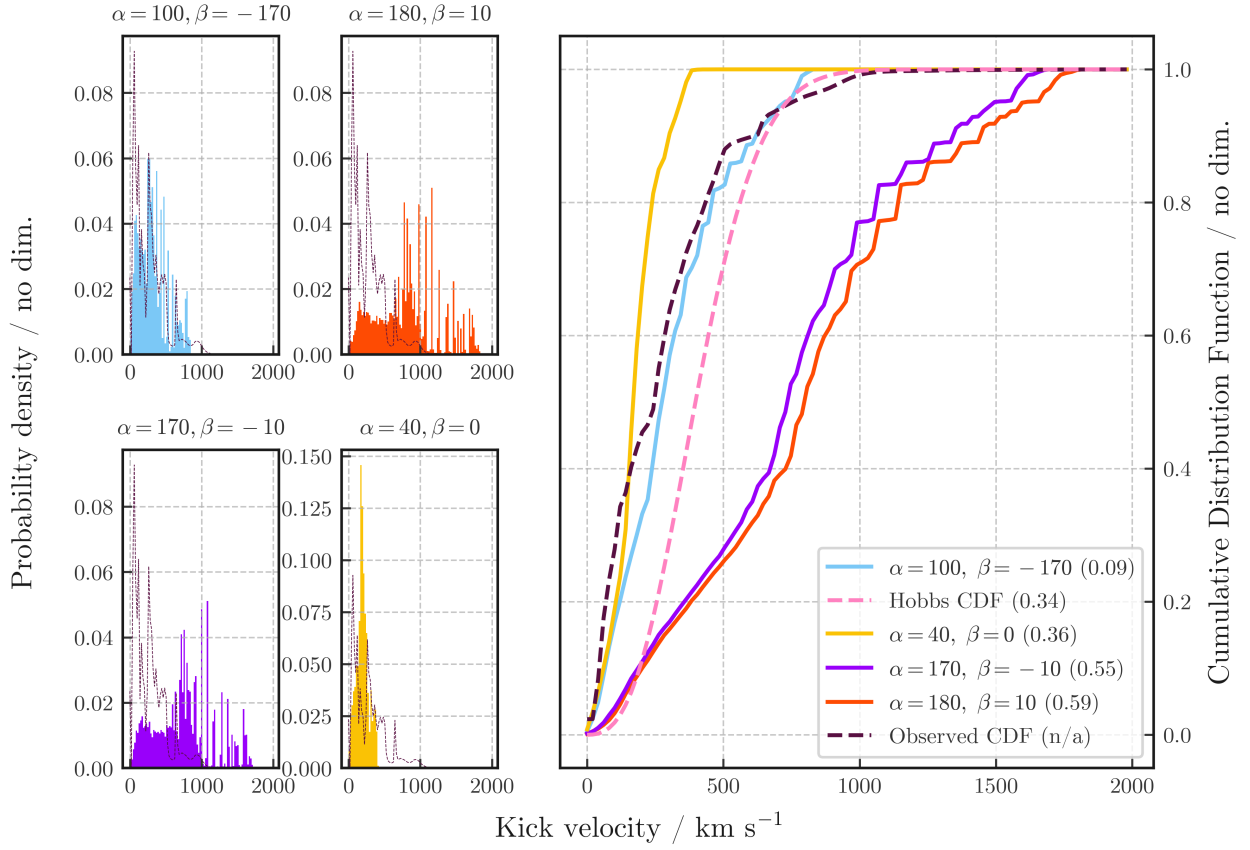


Figure 4.9: The PDFs and CDFs of four systems, alongside the Hobbs CDF and Willcox et al. (2021) CDF. The dashed line on the PDF plots is the PDF of the observed distribution from Willcox et al. (2021). On the right hand plot, the number in parentheses represents the KS statistic for the distribution, measured relative to Willcox et al. (2021), which by definition has no KS statistic.

limit.

4.2.4 | Production of Ultra-Stripped Supernovae

As discussed earlier, the fiducial values of the Bray kick from [Bray & Eldridge \(2018\)](#) admit some problematic extremal cases. If $m_{\text{ej}} \rightarrow 0$, as is the case in a USSN, the kick velocity would become negative. For instance, an “average” USSN with $m_{\text{ej}} = 0.15 M_{\odot}$ and $m_{\text{rem}} = 1.45 M_{\odot}$ would have a kick velocity of $\sim -159 \text{ km s}^{-1}$. The concern here is twofold. Firstly, a negative kick velocity would mean that the velocity vector of the kick would be antialigned with the direction of the more massive remnant; this is a violation of conservation of momentum. Secondly, a kick with high magnitude means that, irrespective of the alignment of the kick, the kick itself is strong. As discussed in [Willcox et al. \(2021\)](#), USSNe are associated with weak kicks, peaking at $v_k \sim 30 \text{ km s}^{-1}$. This means that the fiducial Bray kick is unable to correctly account for USSNe.

To account for these supernova kicks, we introduce a new constraint onto our parameters. Following [Tauris et al. \(2015\)](#), we define a USSN as follows:

1. The ejecta mass is in the range $0 \leq m_{\text{ej}} \leq 0.3 M_{\odot}$, admitting the USSN SN2019dge from [Yao et al. \(2020\)](#),
2. The remnant mass is in the range $1.1 \leq m_{\text{rem}} \leq 1.8 M_{\odot}$,
3. The envelope mass $m_{\text{env}} \leq 0.2 M_{\odot}$.

For every combination α, β , we record whether it is capable of producing one USSN. The constraint is that this heuristic is met. This constraint is extremely conservative, insofar as we do not constrain on the prevalence of USSNe in our sample, but rather by whether or not they are possible at all. The reason for the conservatism here is that the rates of Type Ic USSNe are expected to be very low. In the BPASS dataset, they comprise 1.8 per cent of all CCSNe (Stevance, priv. comm.). The paucity of data alongside the paucity of observational evidence suggests a low rate, but since the prevalence is unknown, we place the conservative constraint of at least one USSN to be agnostic. This effectively widens our error bars such that when the USSNe rate is refined further, it will shrink the parameter space.

4.2.5 | Refinement of the parameter space

We are now in a position to impose constraints on our parameter space based on the analysis provided above. We apply the following four constraints:

1. **DNS Merger Rate** (Section 4.2.1): Does the merger rate produced by these parameters sit along the LIGO contour?
2. **DNS Period-Eccentricity Distributions** (Section 4.2.2): Is $\ln \mathcal{K}_{\alpha\beta} \geq 3.4$ compared to the [Hobbs et al. \(2005\)](#) kick (representing conclusive evidence in favour of the distribution)?

3. **Isolated Pulsar Velocities** (Section 4.2.3): Is the KS statistic produced by this choice of parameters, in comparison to the [Willcox et al. \(2021\)](#) kick, less than 0.5 (indicating it agrees with observational data more than it disagrees)?
4. **Logical USSN velocities** (Section 4.2.4): Is this choice of parameters able to produce at least one USSN?

These can be overlaid onto our parameter space, which is shown in Figure 4.10. The center of mass of our fit region is located at $\alpha = 117^{+36}_{-51} \text{ km s}^{-1}$, $\beta = 13^{+8}_{-14} \text{ km s}^{-1}$, however this value is computed on a finer grid than the parameter space and is thus likely to be an overfit to the data. We thus report the final recommended model value to the nearest half-bin-width of the parameter space, broadening error bars where necessary, and find $\alpha = 115^{+40}_{-55} \text{ km s}^{-1}$, $\beta = 15^{+10}_{-15} \text{ km s}^{-1}$ as the recommended model value. Error bars are always broadened here; traditional rounding would result in the upper bound on α (being 153 km s^{-1}) becoming 150 km s^{-1} , which artificially removes potentially permissible regions of the parameter space. Broadening the bars removes this at the cost of potentially admitting values that do not belong within the “acceptable” zone. Further constraints, however, will shrink the region, so this is an acceptable tradeoff.

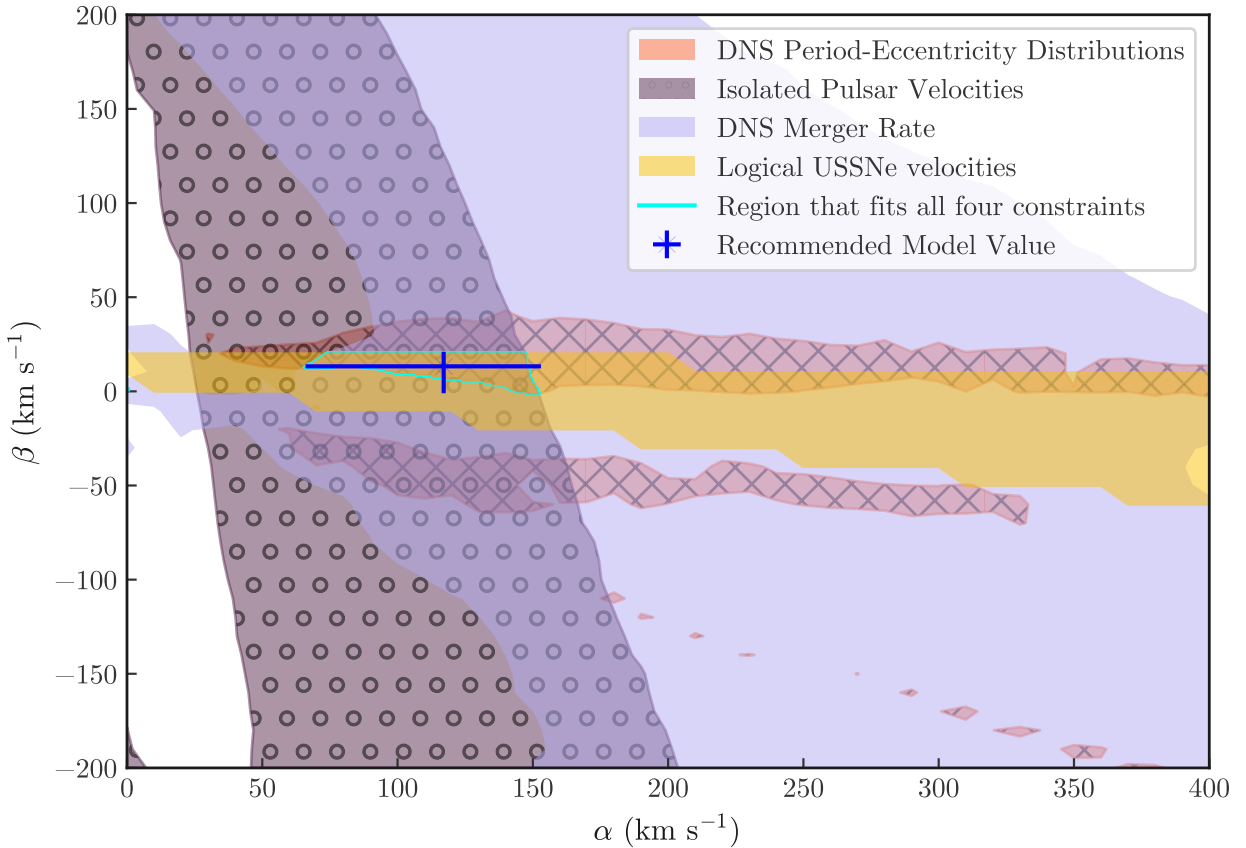


Figure 4.10: The four constraints we apply overlaid on our FINEGRID parameter space. The cyan region represents the region which fits all four parameters, and the blue cross represents the “center of mass” of this region.

In Figure 4.11, we present the two major inferred properties of our recommended model value. First, we present the CDFs of a population governed by a Bray kick with the recommended parameters, contrasted

with the associated Hobbs kick and observed Willcox et al. (2021) data. We also show the period-eccentricity distribution of the population, with the observed systems from Table 4.2 overlaid atop the distribution.

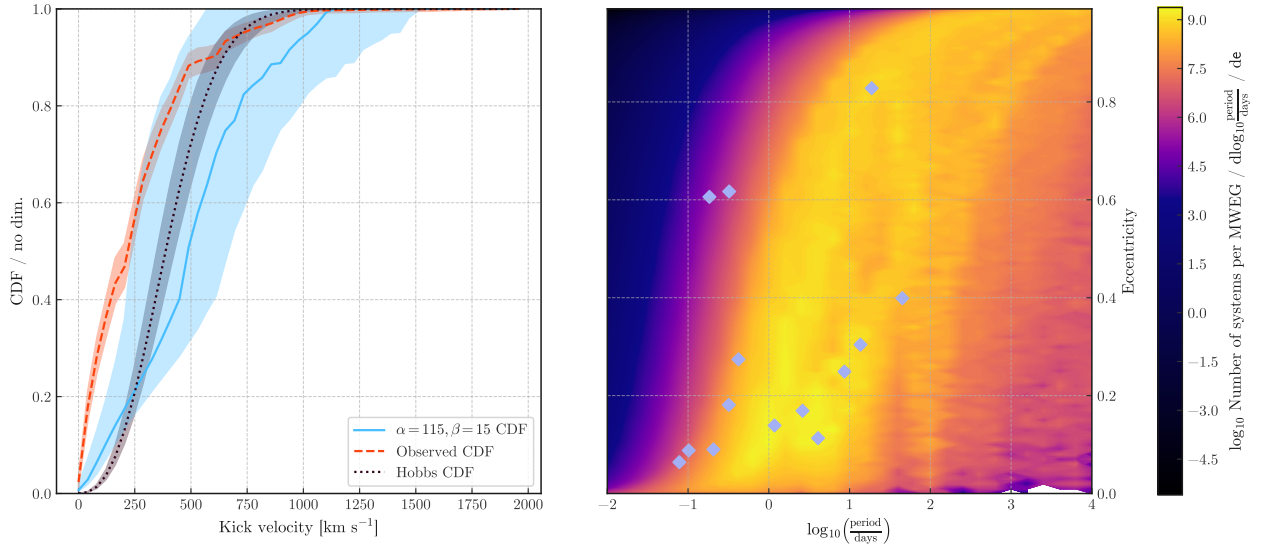


Figure 4.11: The CDF for a population of isolated pulsars (left) and the period-eccentricity distribution (right) for $\alpha = 115 \text{ km s}^{-1}$, $\beta = 15 \text{ km s}^{-1}$. The blue region on the CDF plot shows the uncertainty region for the reported value. The red region shows the uncertainty band on the Willcox et al. (2021) CDF, which was computed by sampling the PDF 10,000 times, taking 2,000 samples each time, and plotting the extent of the simulated distribution. The black band is the uncertainty on the Hobbs distribution, taking $\sigma_{\text{max}} = 291 \text{ km s}^{-1}$ and $\sigma_{\text{min}} = 239 \text{ km s}^{-1}$ from Verbunt et al. (2017).

We are also able to extrapolate this to comment on the expected peak of the event-rate distribution. Given that the event rate is a convolution between the merger rate of our population and the SFRD we choose (cf. Eqn(2.34)), by affixing the SFRD across our populations, by Eqn(2.30) we can vary $\mathcal{R}(t)$ by varying α and β . Since the SFRD is of a Gaussian form (with a peak at $z \sim 1.863$), we can also extract the peak of the distribution assuming it is governed by a particular α - β combination. The general results across the parameter space are shown in Figure 4.12. The best model choice parameters we find fall within the valley of $z = 0$, indicating that the peak of the event rate distribution is the present day.

We note that there is a single peak at $\alpha = 330 \text{ km s}^{-1}$, $\beta = -70 \text{ km s}^{-1}$ which appears to be an outlier when compared to its neighbours. We display in Figure 4.13 the local behaviour of the 9 cells around this region. One can see that this beak is bimodal with a second peak at $z = 0$, which is only 0.098 per cent lower than the reported peak, clearly resulting in a simple outlier.

Unfortunately, there is a paucity of observational evidence to confirm whether or not the peak is truly the present day. Given that a redshift of $z = 2$ corresponds to a lookback time of $t_{\text{LB}} = 10 \text{ Gyr}$, we lack evidence to corroborate this prediction. The next generation of telescopes, including JWST, may yet be able to constrain this result further. In the interim, indirect evidence may be able to be applied via the r-process enrichment history of the Milky Way. Van de Voort et al. (2022), for example, finds that greater velocity natal kicks would be correlated with fewer r-process enhanced stars in the Milky Way, as higher kicks would unbind more binaries, meaning that r-process elements are no longer able to accrete onto star-forming regions. Kobayashi et al. (2023) indicates that current population synthesis models⁶ are unable to reproduce the abundance

⁶Explicitly considered in the paper are COMPAS (Team COMPAS: Riley et al., 2022), StarTrack (Belczynski et al., 2020),

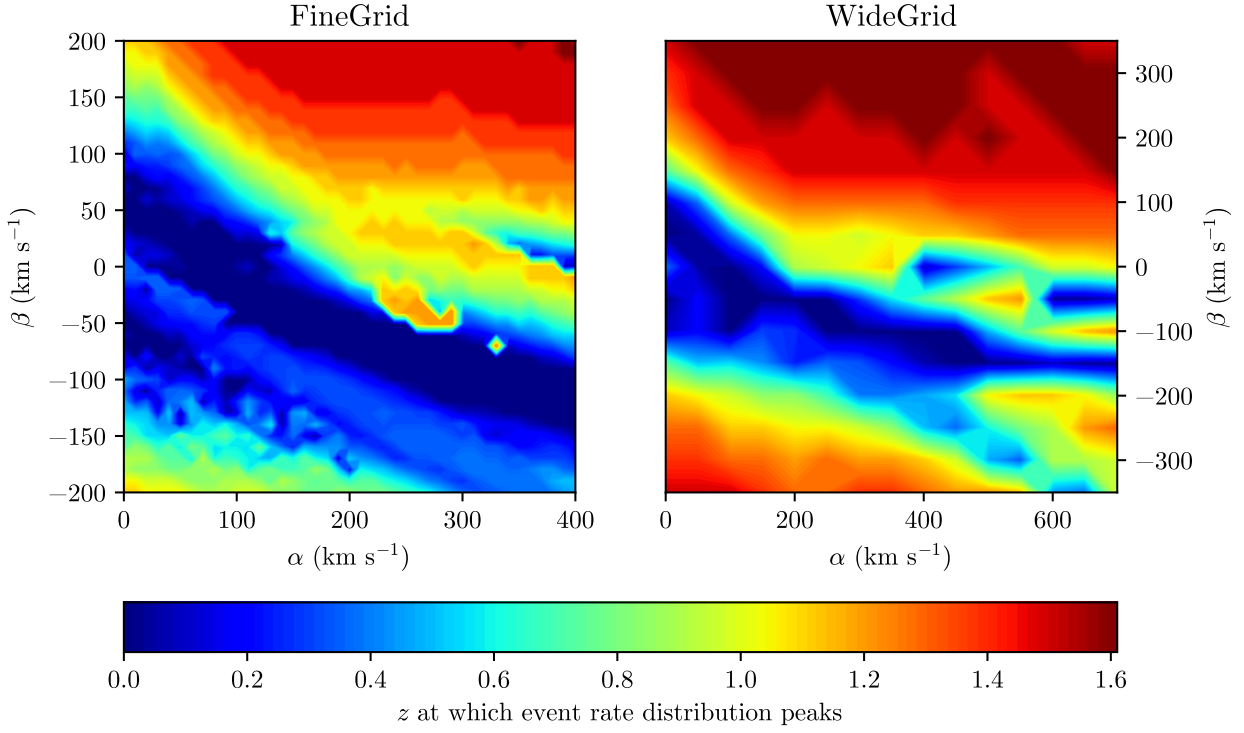


Figure 4.12: The peak of the event rate $\mathcal{R}(t)$ as a function of α and β . Varying these two parameters generally brings the peak closer to the present day ($z = 0$). Absent a merger rate (or, more formally, assuming the merger rate is instantaneous and given by a Dirac-delta function), the SFRD peaks at $z \sim 1.863$. The region surrounding the peak at $\alpha = 330, \beta = -70$ is shown in Figure 4.13.

distributions of certain high-mass isotopes, including the abundances of Oxygen, Iron, Europium, Gold, and Uranium.

4.3 | Discussion

There are limitations to this analysis. The first is that we did not preferentially weight our binaries. [Tauris et al. \(2017\)](#) for example places particular weight on PSR J1930-1852, which is a particularly wide binary. Per [Swiggum et al. \(2015\)](#) (who reported it), “its unique spin and orbital parameters challenge models that describe BNS formation”, which implies that any credible model of BNS formation should replicate its orbital parameters. We did find in Figure C.1 that some parameters had non-zero probability of replicating this pulsar, which indicates that the Bray kick is able to reproduce it.

Secondly, we have assumed Newtonian (specifically, Keplerian) orbits. Formally, the derivations presented earlier (cf. Section 2.4.1) assumed general relativity for the existence of the gravitational wave equation,⁷

$$\square \bar{h}_{\mu\nu} = -\frac{1}{2}\sqrt{32\pi G}T_{\mu\nu}, \quad (4.9)$$

but the injected physics was Newtonian (see Section A.3 for a derivation). Strictly speaking, relativistic corrections apply to these kinematic equations too, though these only become dominant when the two stars are

Brussels ([Mennekens & Vanbeveren, 2014, 2016](#)), ComBinE ([Kruckow et al., 2018](#)), and BPASS ([Eldridge et al., 2017](#)).

⁷Here, $\bar{h}_{\mu\nu}$ is the trace inverse of a weak perturbation to locally Minkowski spacetime with positive-definite trace, and $\square$ is the d’Alembertian operator.

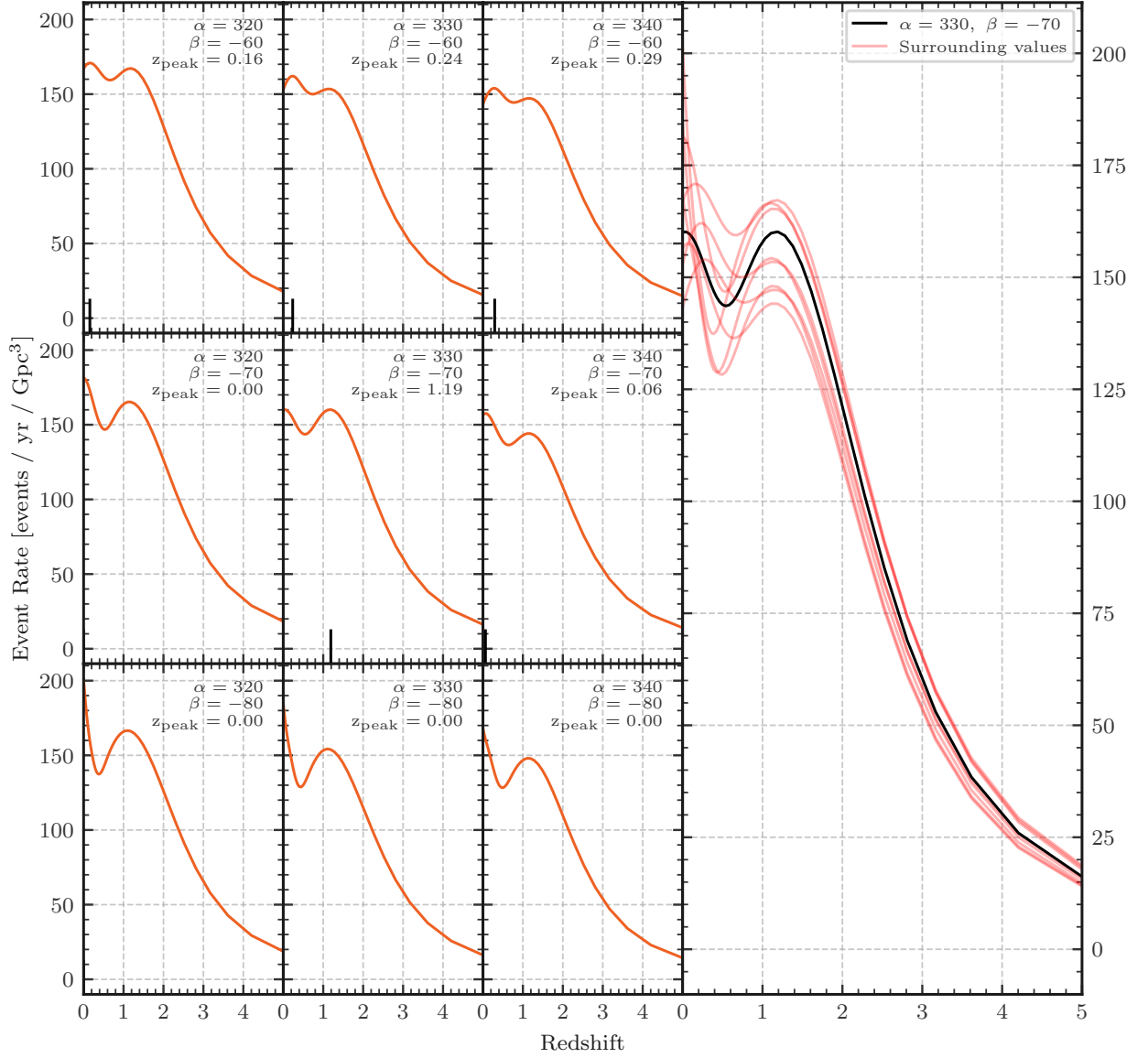


Figure 4.13: The event rate $\mathcal{R}(z)$ for a region surrounding $\alpha = 330, \beta = -70$ on FINEGRID. This region corresponds to the “outlier” in Figure 4.12. Shown on the left are the outlier bin in the center and its neighbours in both α -space and β -space. The black line on each plot denotes the location in redshift of the event rate distribution peak. The right plot shows the outlier in black and its neighbours in faded red behind.

closer than the innermost stable circular orbit (ISCO), given by $r_{\text{isco}} = \frac{6GM}{c^2}$, where matter can be exchanged. To account for this, one would need to apply corrections to the two-body equation of motion, termed the post-Newtonian approximation. The general equation becomes

$$\frac{d\vec{v}}{dt} = \frac{G(m_1 + m_2)}{r^2} \left(-\hat{n} + \sum_{k=2}^N \frac{1}{c^k} \vec{A}_{\frac{k}{2}\text{PN}} \right), \quad (4.10)$$

where N is the highest order term considered, and $\vec{A}_{\frac{k}{2}\text{PN}}$ is termed the “ $\frac{k}{2}$ -th Post-Newtonian” coefficient. Testing with the RIRORIRO code from [van Zeist et al. \(2021\)](#) indicated that including terms up to $N = 6$ (3PN) increases the merger time by fractions of a year, and thus does not significantly alter the merger rate density we observe. Typically 3.5PN ($N = 7$) is used as an upper bound ([Itoh & Futamase, 2003](#)), though up to 10PN ($N = 20$) has been used before ([Blanchet, 2014](#), and references therein).

Relativistic corrections also typically manifest in using more sophisticated spacetime metrics. Eqn(4.9) assumes a metric which is locally Minkowski, that is

$$g_{\mu\nu} = \eta_{\mu\nu} + \sqrt{32\pi G} h_{\mu\nu}, \quad (4.11)$$

where $\eta_{\mu\nu}$ is the positive-definite Minkowski metric. This “weak-field” approximation breaks down when a compact object enters within its companion’s ISCO. Given gravitational waves result in strains on the order of 10^{-19} ([van Zeist et al., 2023](#)), representing observed fractional shifts on Earth smaller than the diameter of a proton, ([Moore et al., 2014](#)), the weak field regime suits, and again modelling the relativistic corrections to the post-ISCO regime only increases the merger rate by fractions of years.

To contextualise this case study, other authors such as [Mapelli & Giacobbo \(2018\)](#); [Wiktorowicz et al. \(2019\)](#); [Giacobbo & Mapelli \(2020\)](#); [Santoliquido et al. \(2021\)](#); [Kapil et al. \(2023\)](#) have explored the parameter space of BNS kicks, though none have performed the analysis in the way we have. These works typically vary the progenitor population properties, in such a way that the formed binaries can in principle vary strongly. However, in our population synthesis procedure, we are able to effectively affix the physics between different models such that *only* the kick velocities are expected to vary between individual populations. Since the Bray kick has two parameters which can be altered, we are able to modify the kick velocity without changing the physics, thus allowing us to isolate and study the kick mechanism alone.

We found a selection of 13 different pairs of parameters that fit our four constraints. For values of $\alpha \geq 100 \text{ km s}^{-1}$, we found that low magnitude, but non-zero, values of β replicated the BNS merger rate from [LIGO-Virgo-KAGRA \(2021\)](#) and replicating the observed BNS systems compiled in [Vigna-Gómez et al. \(2018\)](#). A true ‘best’ set of parameters likely lie within this subsample of our parameter space.

Whilst the current observed BNS merger rate is limited, and has wide uncertainty bands, further observations will allow the distribution of the merger rate over redshift to become more refined. Since the peak of the event rate is dependent on the DTD, this distribution is also sensitive to the natal kick and thus can be used as a further constraint on the Bray kick parameters. In Figure 4.12 we display the location in redshift of this peak for our FINEGRID and WIDEGRID model sets. We find that the peak redshift across both grids

corresponds to $z < 1.5$, with a very clear and defined valley of the present day. Across the 13 pairs of parameters that fulfil all of our constraints, we also find that the peak corresponds to $z \leq 0.26$, and the best model $\alpha = 115 \text{ km s}^{-1}, \beta = 15 \text{ km s}^{-1}$ peaks in the present-day, at $z = 0$.

We have also highlighted the caveat that $\alpha = 115^{+40}_{-55} \text{ km s}^{-1}, \beta = 15^{+10}_{-15} \text{ km s}^{-1}$ should not be interpreted as ‘the correct choice of parameters’ but rather ‘the mean of the set of parameters which replicates the four observables we test against’. It is also pertinent to note that these values are calibrated against the BPASS version 2.2.1 models (and should not change significantly when used with future stellar models from the same stellar evolution code). Using these values with other BPS codes should be approached with caution, and readers using other codes should calibrate their choice of parameters against their BPS codes in the same way that we have.

As part of the calibration and selection of binary parameters, we simulated 1,000 kicks per binary. Testing showed a slight, but not unexpected, dependence on number of simulated kicks. We found that adjusting the number of kicks simulated by a factor of ten resulted in a shift of $\log_e \mathcal{K}$ by approximately one.

Authors such as [Fryer et al. \(1998\)](#); [Arzoumanian et al. \(2002\)](#); [Bombaci & Popov \(2004\)](#) suggest that the kick distribution should ostensibly contain two peaks. In testing, we found no combination of α and β for the Bray kick produced an obviously bimodal single-star velocity distribution, though we do observe interesting structure in the resultant PDFs. Figure 4.11 clearly shows several changes of gradient in the CDF, which can be interpreted as several ranges of kick behaviour. This behaviour arises because the Bray kick is linked to the ejecta mass distribution. Thus the behaviour of low/high kick populations could be due to the results of stellar evolution causing low/high ejecta masses for CCSNe that form neutron stars.

The reason we did not include a bimodal Hobbs kick in this study is twofold. Firstly, the unimodal Hobbs kick has been extensively discussed in literature (see, among others, [Nakar, 2007](#); [Belczynski et al., 2008](#); [Lorimer, 2008](#); [Dominik et al., 2012](#); [Bray & Eldridge, 2016](#); [Eldridge et al., 2017](#); [Bray & Eldridge, 2018](#)) and thus was a good starting point for our investigation. Secondly, as a point of consideration, we computed the log-Likelihood of reproducing our BNS catalog against a weighted model from [Verbunt et al. \(2017\)](#), taking $w = 0.42, \sigma_1 = 75 \text{ km s}^{-1}, \sigma_2 = 316 \text{ km s}^{-1}$. We found that $\log_e \mathcal{K} = 0.8$. Thus, if this value were to be used as the reference model for the study, it would shift our log-Likelihoods by approximately one order of magnitude and would only admit an extra handful of systems.

In the future, an additional prior can be used on our dataset. The natural interpretation of the Bray kick expression is that β is the ‘limiting value’ that the kick velocity should approach as the ejecta mass gets smaller (as in the case of USSNe and ECSNe). In events such as the direct collapse of a star to a black hole in a failed SNe, where the ejecta mass is zero, β would identically be the kick velocity. Thus, in order for conservation of momentum to be respected for low-ejecta mass kicks, we expect $|\beta| > 0$ – i.e., we expect the magnitude of β to be low, but most likely non-zero. This prior was not imposed on our parameter space – the $\log_e \mathcal{K}$ and USSNe constraints independently restricted β to be > 0 . It is pertinent to note, as well, that even in the limit as the ejecta mass goes to zero, a value of $\beta < 0$ would imply that the velocity vector is anti-aligned with the chosen direction of the kick. $\beta < 0$ also allows for a zero kick velocity when the ejecta

mass is non-zero. At this stage, this scenario cannot be ruled out, thus we cannot yet include this constraint as a prior.

In addition, we found that the mean pair of parameters – the center of the cyan region in Figure 4.10 – corresponded to $\alpha = 115_{-55}^{+40} \text{ km s}^{-1}$, $\beta = 15_{-15}^{+10} \text{ km s}^{-1}$. Although this work does not purport to claim a single ‘best’ pair of parameters, these ones can be interpreted as the pair that optimises each of the individual tests that we have performed. We note that this value lies off-grid, and hence in Figure 4.11 we generated a bespoke data file for this set of two parameters. The optimal form of the kick, for the BPASS models, is therefore

$$v_{\text{kick}} = 115_{-55}^{+40} \left(\frac{m_{\text{ejecta}}}{m_{\text{remnant}}} \right) + 15_{-15}^{+10} \text{ km s}^{-1}. \quad (4.12)$$

In sum, the key conclusions we draw are:

1. Providing multiple constraints allows for the parameter space of possible kick configurations to be drastically reduced.
2. Admissibility of kick velocities of $\lesssim 30 \text{ km s}^{-1}$ for USSNe is one of the tightest constraints we can provide.
3. The peak of the merger rate distribution is likely to be the present-day, and this is a further constraint we can apply to the dataset when more information is available.
4. The center of mass of the permitted region of the Bray parameter space is $\alpha = 115_{-55}^{+40} \text{ km s}^{-1}$, $\beta = 15_{-15}^{+10} \text{ km s}^{-1}$, although caution should be exercised before using this value without calibration.

The analysis presented here is an analysis pipeline to refine the properties of kick models against four different observables. In principle, it can be applied to any kick model to validate the model. Our statistical analysis provides a simple way to validate kick models: it can be used to rule out either magnitudes of kick velocities or entire physical kick prescriptions.

4.4 | Chapter Summary

We have presented a formalism for evaluating and quantifying the goodness-of-fit of the natal kick mechanism against four independent tests. The tests we have presented – constraining the merger rate of BNS systems, the period-eccentricity distribution of Galactic BNS systems, the kick velocities of single-star pulsars, and the production of ultra-stripped supernovae – greatly constrain the parameter space to be less than 1 per cent of the possible parameters. Furthermore, the data suggest that the peak of the merger rate distribution is the present day, with $z \simeq 1.5$ as the upper bound of the entire parameter space, and $z = 0$ for our recommended value.